DIPLOMA PROGRAMME PRACTICE · HL · SET B

Chemistry / Paper 1B

Question booklet · 35 marks · 120 minutes across 1A and 1B

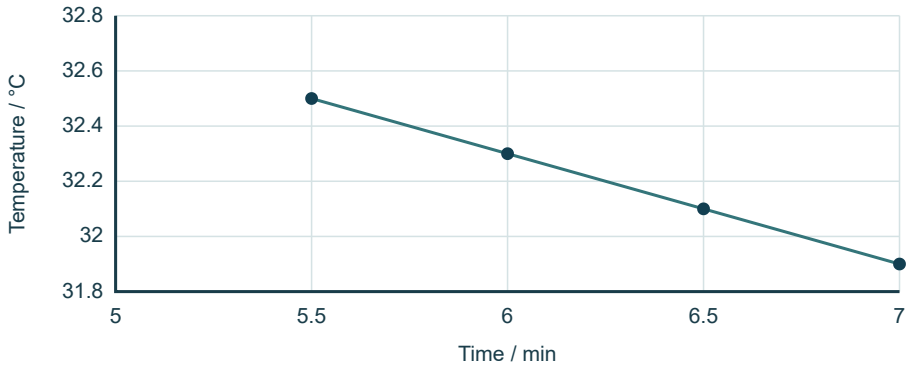
Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [12]

Correcting a combustion measurement

A burner heats water in a vessel with independently measured heat capacity 62.0 J K^{-1}. Water mass is 180.0 g; use c=4.18 J g^{-1} K^{-1}. Burning stops at t=5.0 min. The initial water temperature was 19.4 °C and the fuel mass loss is 0.620 g. Fuel molar mass is 60.1 g mol^{-1}.

Cooling time / min	5.5	6.0	6.5	7.0
Temperature / °C	32.5	32.3	32.1	31.9



Temperature / °C against Time / min. Data points are shown by filled circles.

(a) Use the cooling trend to estimate the temperature at shutdown and the corrected temperature rise. **[3]**

(b) Calculate the energy absorbed by water and vessel, and the experimental molar enthalpy of combustion. **[3]**

(c) A reference value for the same fuel and product states is −2020 kJ mol^{-1}. Calculate the percentage of reference energy recovered and explain one chemical and one physical source of deficit. **[3]**

Paper 1B · Question 1 (continued)

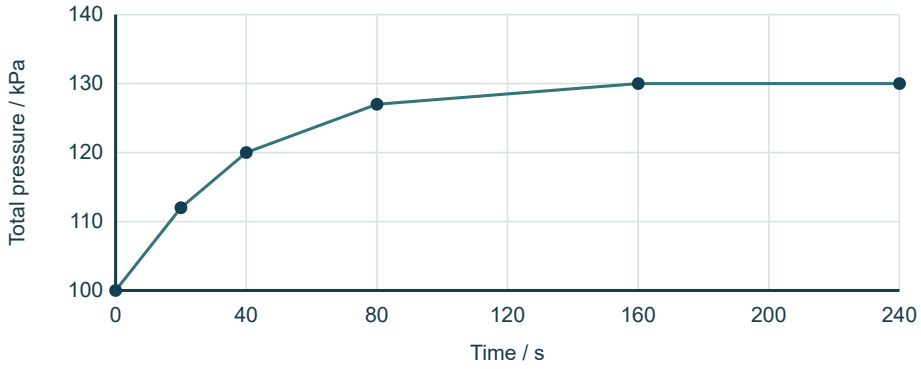
(d) Suppose fuel evaporates unburnt during weighing. Predict its effect and propose a paired control to estimate its magnitude. **[3]**

QUESTION 2 [12]

Pressure tracking in a sealed cell

Initially a rigid 1.000 dm³ vessel contains only $N_2O_4(g)$ at 300.0 K and 100.0 kPa. It is held at 300.0 K as $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ approaches equilibrium. Treat gases as ideal, with R=8.31 kPa dm³ mol⁻¹ K⁻¹. The final pressure is 130.0 kPa.

t / s	0	20	40	80	160	240
p / kPa	100	112	120	127	130	130



Total pressure / kPa against Time / s. Data points are shown by filled circles.

(a) Determine the initial amount of N_2O_4 and the fraction that dissociates. **[3]**

(b) Calculate both equilibrium concentrations and Kc using concentrations in mol dm⁻³. **[3]**

(c) Estimate the mean formation rate of NO_2 from 20 to 80 s. Explain the factor linking increase in total moles to moles of NO_2 formed. **[3]**

(d) A student interprets a rising pressure as sufficient proof of dissociation. Give a competing explanation and an experimental check. Explain why a plateau alone does not prove dynamic equilibrium. **[3]**

QUESTION 3 [11]

A cell under load

A Zn/Zn^{2+} and Cu^{2+}/Cu cell is measured at 298 K. Standard reduction potentials are −0.76 V and +0.34 V respectively. Both initial solutions are taken as standard for the theoretical calculation. Use F=96,500 C mol^{-1}. The meter's absolute uncertainty is ±0.01 V for each reading.

Condition	Fresh, open circuit	Connected to resistor	Open circuit after disconnecting
Voltage / V	1.09	0.93	1.08

(a) Write the net ionic reaction and calculate the standard potential and Gibbs energy change per mole of Zn. **[3]**

(b) Compare the two open-circuit readings with each other using maximum absolute uncertainties. Is the decline resolved? **[3]**

(c) Explain why the loaded voltage is unsuitable for a direct calculation of standard Gibbs energy and identify the direction of electron flow through the resistor. **[2]**

(d) A technician replaces the salt bridge with a dry insulating strip. Explain the resulting behaviour and propose a control that tests the bridge rather than the electrodes. **[3]**